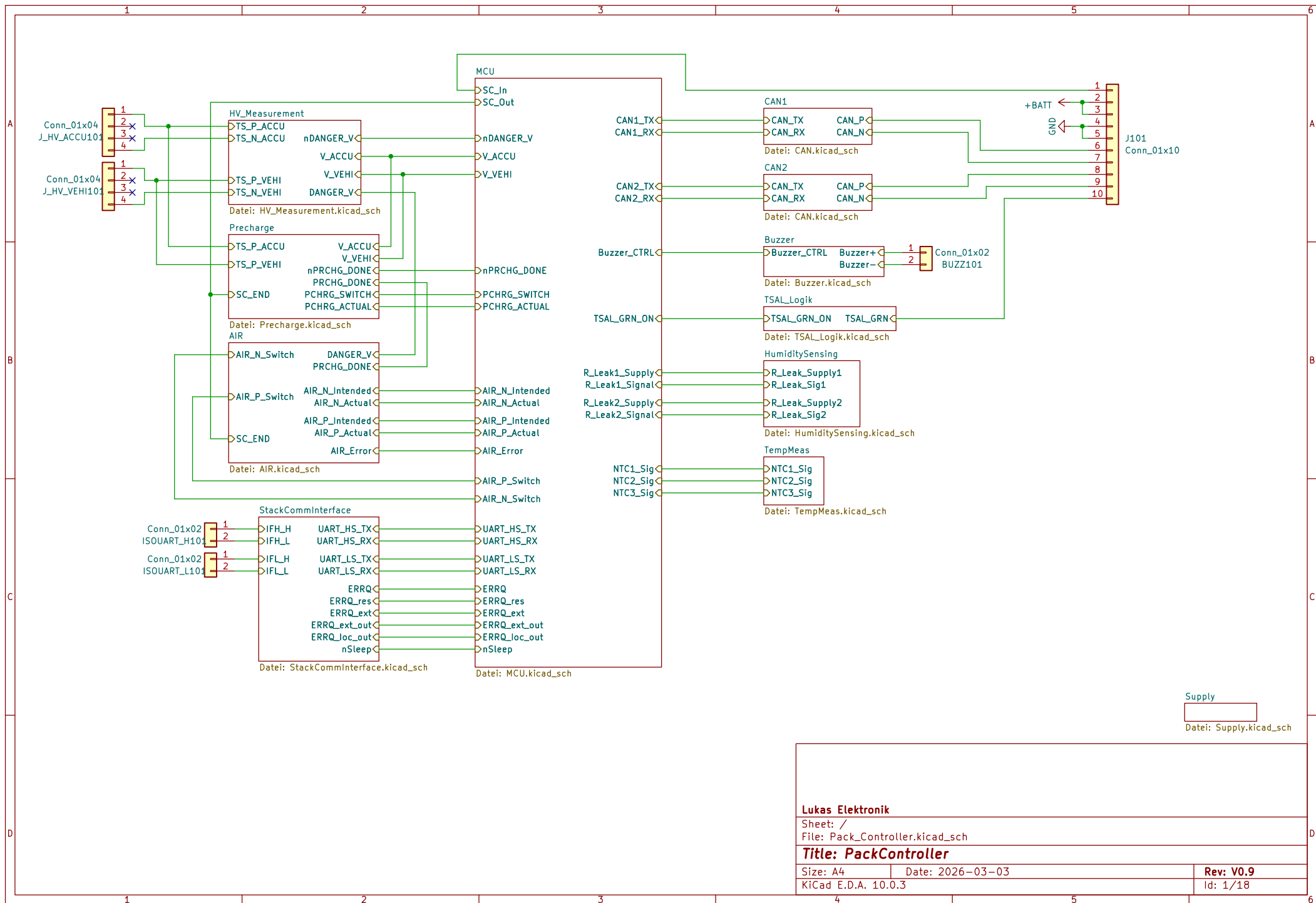
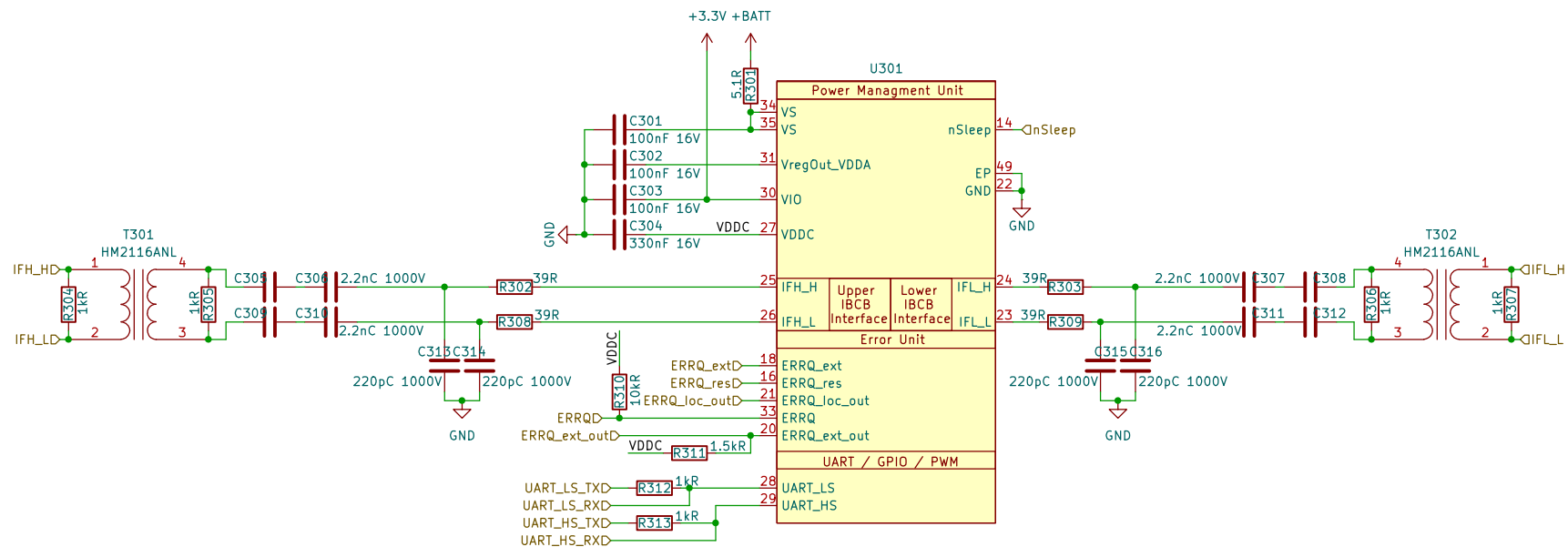






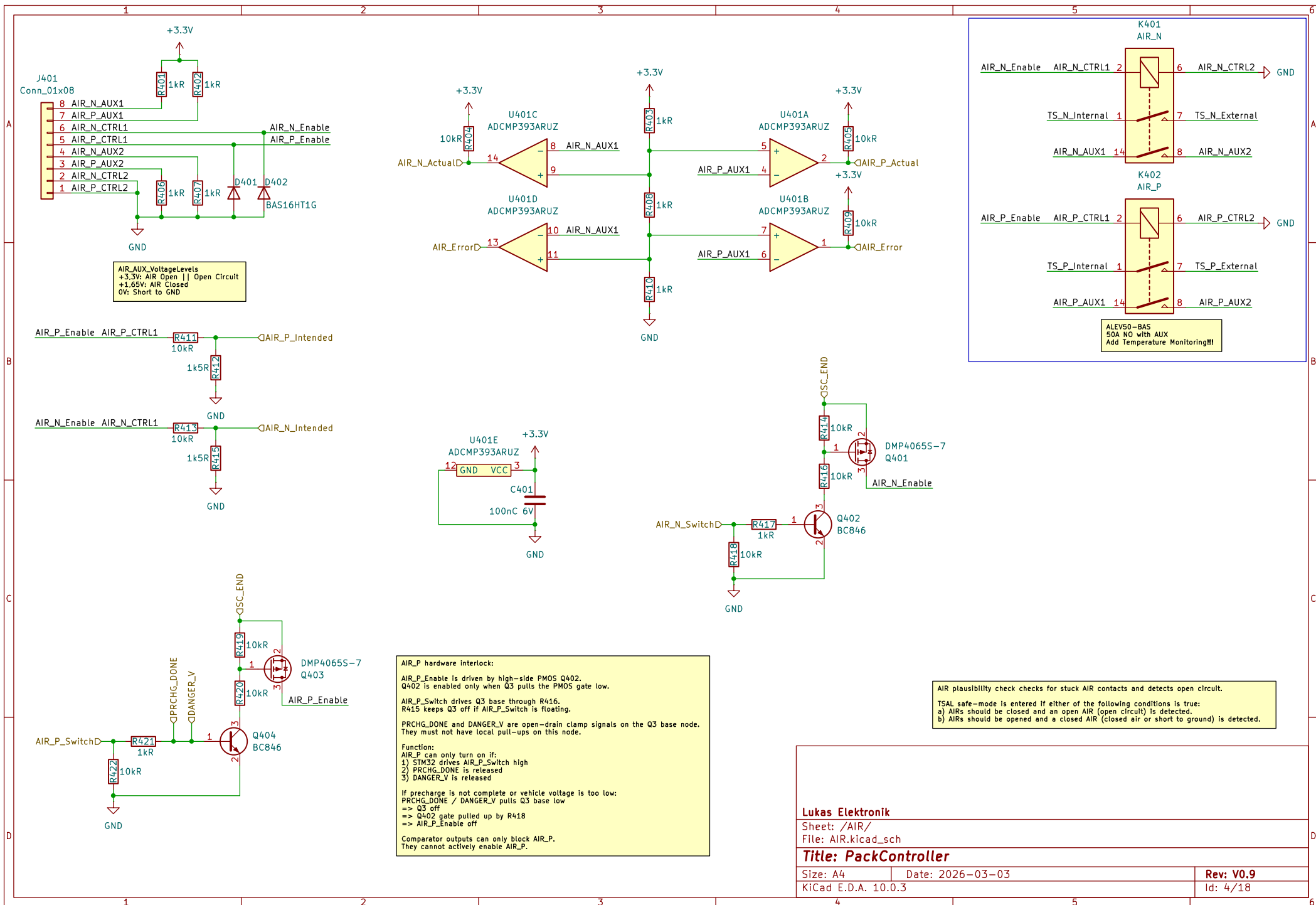
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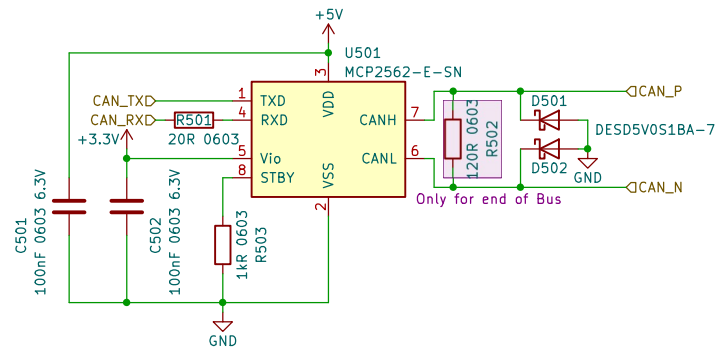
Sheet: /StackCommInterface/
File: StackCommInterface.kicad_sch

Title: PackController

Size: A4 Date: 2026-03-03
KiCad E.D.A. 10.0.3

Rev: V0.9
Id: 3/18





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Sheet: /CAN1/
File: CAN.kicad_sch

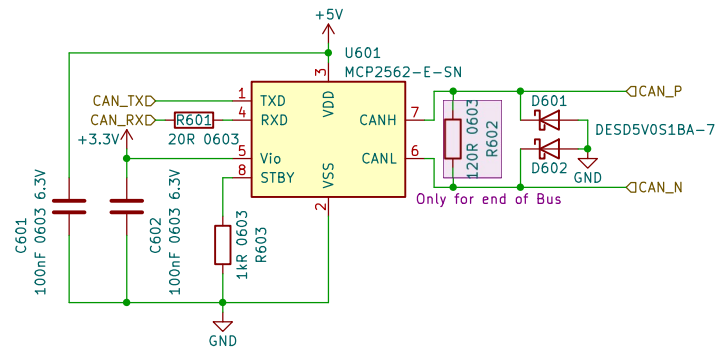
Title: PackController

Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 5/18



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Sheet: /CAN2/
File: CAN.kicad_sch

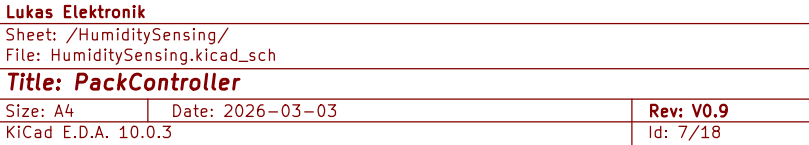
Title: PackController

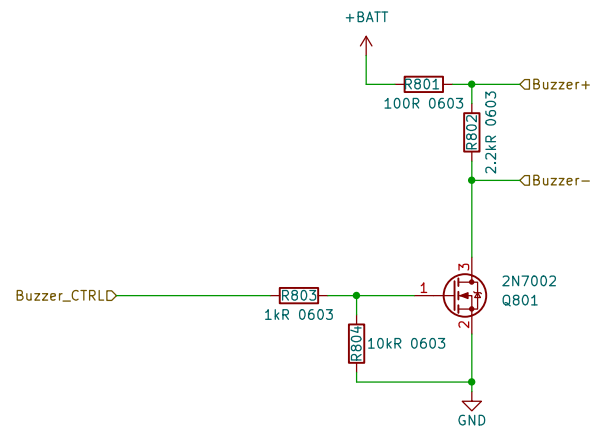
Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 6/18





Passive piezo buzzer driver.
Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
 $f_{PWM} \approx 4 \text{ kHz}$
duty $\approx 50 \%$

Q15 is a low-side NMOS switch.
R37 discharges the piezo during MOSFET off-time:
 $\tau = R37 * C_{PIEZO} \approx 2.2k * 24nF \approx 53\mu s$.

R36 limits peak current and EMI.
No additional series coupling capacitor required.

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Sheet: /Buzzer/

File: Buzzer.kicad_sch

Title: PackController

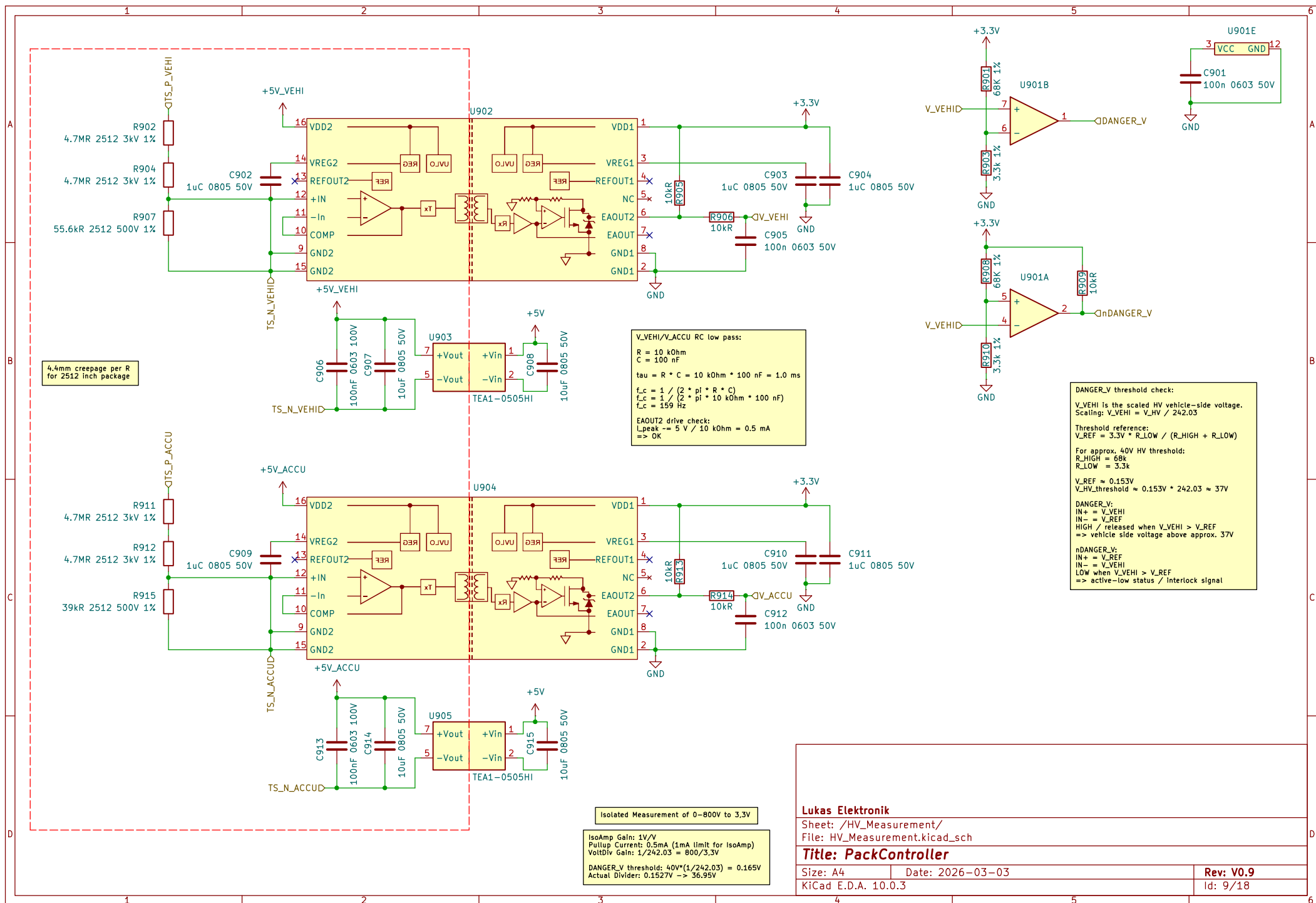
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 8/18



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Sheet: /HV_Measurement/

File: HV_Measurement.kicad_sch

Title: PackController

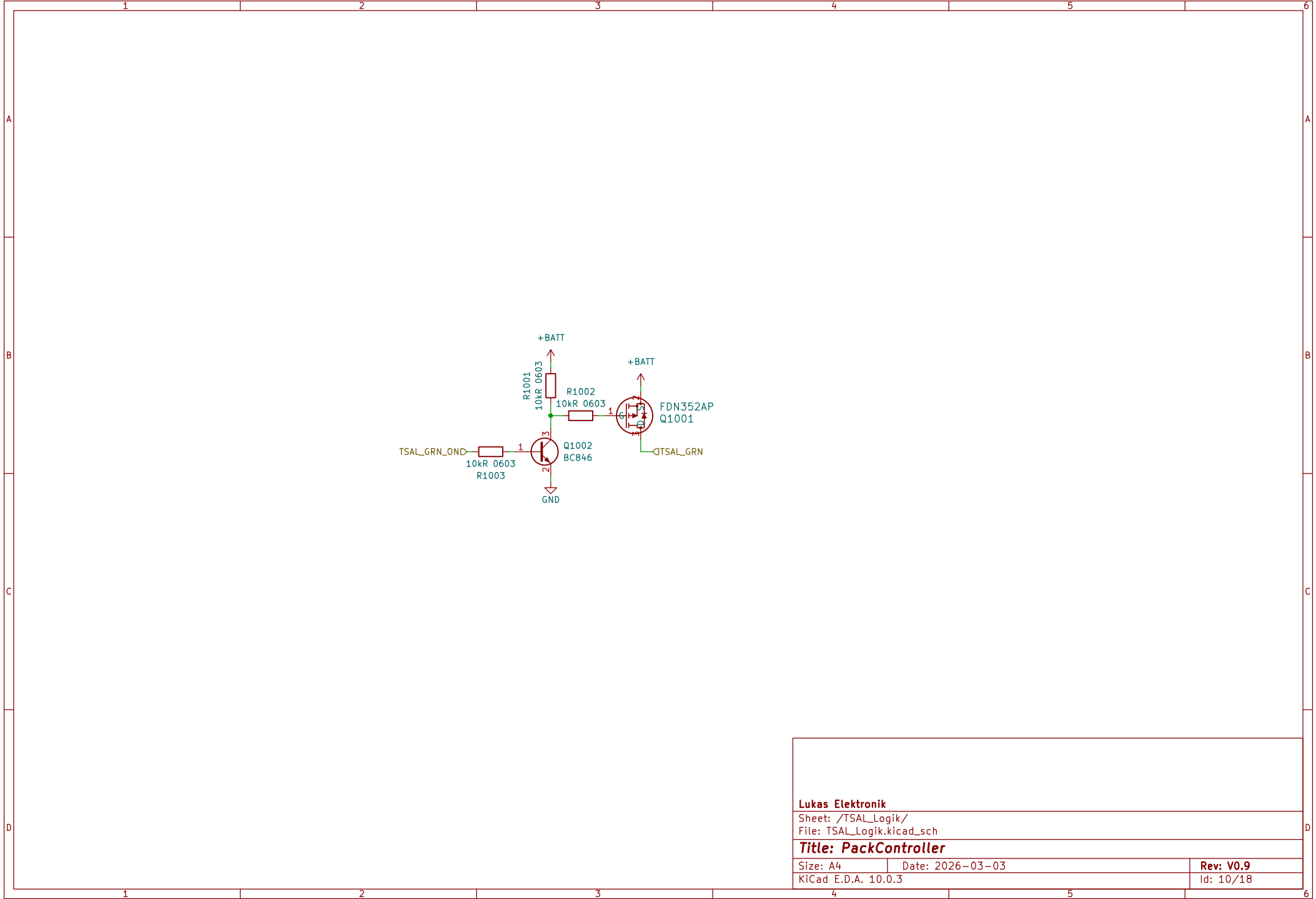
Size: A4

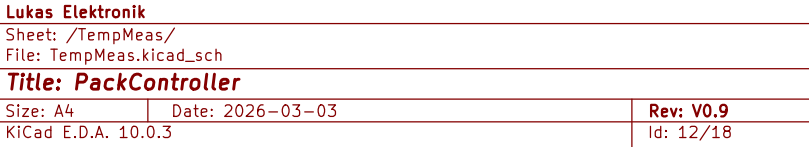
Date: 2026-03-03

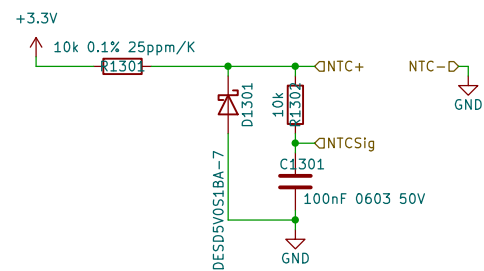
KiCad E.D.A. 10.0.3

Rev: V0.9

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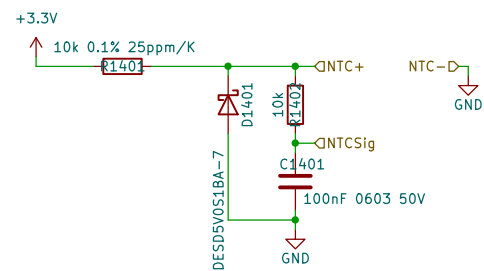
External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$R_{NTC} = R_{PULLUP} * x / (1 - x)$
 $x = ADC_{raw} / ADC_{max}$

R58/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC1/		
File: NTC1.kicad_sch		
Title: PackController		
Size: A4	Date: 2026-03-03	Rev: V0.9
KiCad E.D.A. 10.0.3		Id: 13/18



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC2/
File: NTC1.kicad_sch

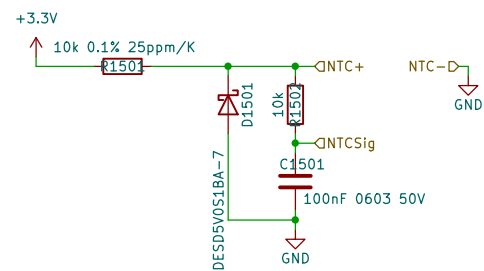
Title: PackController

Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

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External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$R_{NTC} = R_{PULLUP} * x / (1 - x)$
 $x = ADC_{raw} / ADC_{max}$

R5B/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC3/

File: NTC1.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

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Id: 15/18

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open-wire detection.
 $R_SENSOR = R_LEAK || R_EOL$

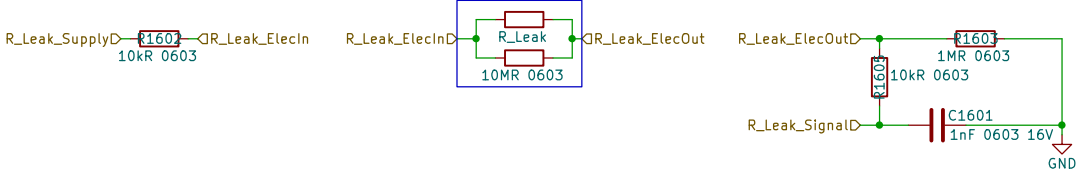
Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$
 $R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC ≈ 0
dry connected: R_SENSOR ≈ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k



Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open-wire detection.

$R_SENSOR = R_LEAK || R_EOL$

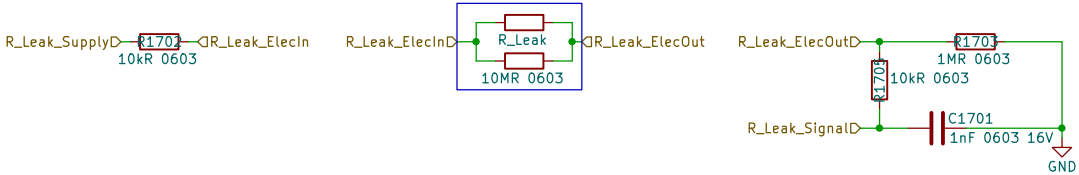
Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$
 $R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC $\approx$ 0
dry connected: R_SENSOR $\approx$ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k

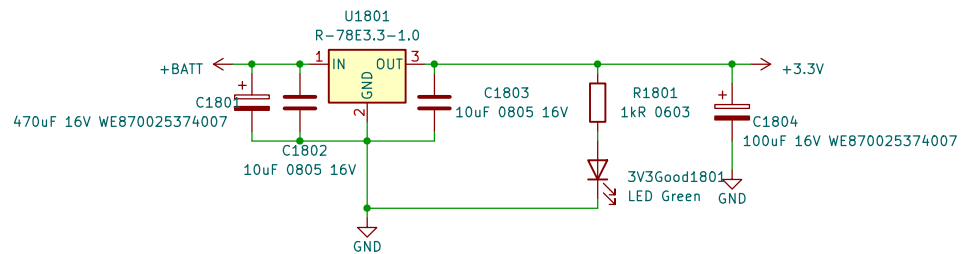


+3V3 Supply worst-case estimate:

Main loads:
STM32G483 active incl. peripherals: -50 mA
ADuM3190 VDD1, 2x: -4 mA
24LC256 EEPROM write: -3 mA
MCP2562 VIO, 2x worst-case: -1 mA
ADCMP393 comparators + logic: <1 mA
Pullups, dividers, GPIO drivers, LEDs: -20...30 mA

Estimated worst-case load: -80...90 mA
Selected supply: RECOM 3.3 V / 1 A DC/DC

Design margin:
1 A / 90 mA ≈ 11x

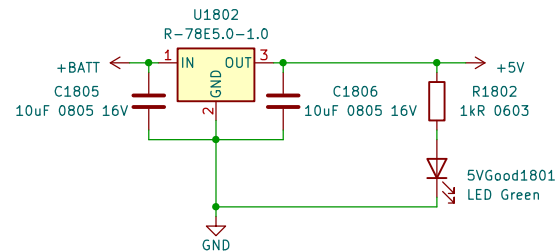


+5V Supply worst-case estimate:

Main loads:
TEA1-0505HI Isolated DC/DC no-load, 3x: -84 mA
TEA1 loads incl. ADuM3190 Isolated sides: -25...30 mA
MCP2562 CAN transceivers, 2x dominant: -140 mA
Misc. reserve / startup / tolerance: -50 mA

Estimated worst-case load: -250...300 mA
Selected supply: RECOM 5 V / 1 A DC/DC

Design margin:
1 A / 300 mA ≈ 3.3x



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Sheet: /Supply/
File: Supply.kicad_sch

Title: PackController

Size: A4 Date: 2026-03-03

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Rev: V0.9

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